

CYBERSECURITY INTERNSHIP TRAINING REPORT

Cybersecurity Fundamentals



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Your roadmap to understanding the fundamentals of cybersecurity.

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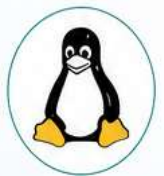


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LEARNING PATH

A step-by-step journey to master cybersecurity.



Learn
(Concepts)



Practice
(Labs)



Analyze
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1 Cybersecurity Basics

- Understand The CIA Triad :
Confidentiality, Integrity, Availability.
- Explore Threat Types :
Phishing, Malware, DDoS, SQL Injection, Brute Force, Ransomware.

Study Attack Vectors :

Social Engineering, Wireless Attacks, Insider Threats.

1 Understand the CIA Triad :

Confidentiality, Integrity, Availability.

- CIA Triad
- Confidentiality
- Availability

- What is CIA Tried :

The CIA tried is one of the most impartment fundamental concepts in cyber security.

CIA stands for Confidentiality, Integrity and Availability.

It provide a basic framework for protecting data,applications,networks and computer system from security threats

1 Confidentiality :

Confidentiality means protecting information from unauthorized access or disclosure

Only authorized users should be able to view sensitive information

Examples :

- Password protection
- Encryption
- User Authentication
- Access control
- Multi factor authentication (MFA)

Example :

In an online banking application, only the account owner and authorized bank employee should be able to access private account information

2 Integrity :

Integrity means ensuring that information remains accurate complete and trustworthy and is not modified and unauthorized users

Examples :

- Hashing
- Digital signatures
- File permissions
- Data validation
- Version control

Example :

If an attackers changes a bank account balance from 10,000 to 1,00,000 without authorization, the integrity of the data has been compromised

3 Availability :

Availability means ensure that authorized users can access data and services when they are needed

Examples :

- Regular backup
- Redundant servers
- Disaster recovery
- Load balancing
- DDoS protection

Example :

If an online banking website becomes unavailable because of a cyber attack, legitimate customers cannot access their accounts. This is a violation of availability

CIA Tried Example

Principle	Meaning	Example
Confidentiality	Prevent unauthorized access	Password + MFA
Integrity	Prevent unauthorized modification	Hashing/digital signatures
Availability	Keep services accessible	Backup servers

2 Explore Threat Types :

Phishing, Malware, DDoS, SQL Injection, Brute Force, and Ransomware.

Threat type

- Phishing
- Malware
- DDoS
- SQL injection
- Brute Force
- Ransomware

[1] Phishing :

Phishing is a type of Social Engineering cyber attack in which an attacker tricks a person into providing sensitive information or performing an unsafe action. The attacker usually pretends to be trusted person, company, bank or online services

How phishing works :

A typical phishing attack follows these steps

Attacker → Fake Email/Message → Victim →
Fake Website/Attachment → Information
Theft

For example victim may receive a message claiming

Your account will be suspended. Click to
verify your account

The link may lead to a fake login page
designed to steal the victim username or
password

Common type of phishing

Email phishing = Fake email designed to steal information

Spear phishing = targeted phishing aimed a specific person or organization

Smishing = phishing through SMS or text messages

Vishing = phishing through phone calls or voice communication

Clone phishing = A legitimate looking copy of a previous message containing a malicious link or attachment

Common warning signs

- Unexpected messages asking for passwords or OTP
- Urgent or threatening language
- Suspicious links or domains
- Unknown attachments
- Requests for confidential information
- Spelling or grammar mistakes

How to Prevent Phishing

- Check the sender carefully.
- Don't click suspicious links.
- Verify the website address before logging in.
- Never share passwords or OTPs.
- Enable **multi-factor authentication (MFA)**.
- Keep browsers and security software updated.
- Report suspicious emails or messages

[2] Malware :

Malware stands for Malicious software.

It is software intentionally created to damage system, steal information, disrupt operations or gain unauthorized access to computers, mobiles, networks or server

How malware works :

A simple malware attack can be represented as:

Attacker → Malware → Victim's Device → Unauthorized Activity

Malware can enter a system through malicious email attachments, unsafe downloads, compromised websites, infected USB devices, or vulnerable software.

(1) Type of malware :

1. Virus :

Attaches itself to legitimate file and can spread when the infected file is executed.

2. Worm

Can spread automatically across networks without requiring the user to manually copy or an infected file.

3 Trojan

Pretends to be legitimate software but performs malicious activities after installation.

4 Ransomware

Encrypts or blocks access to file and demands payment from the victim.

5 Spyware

Secretly monitors user activity and may collect sensitive information

6 Adware

Displays unwanted Ads and some case track user behavior

7 Rootkit

Attempts to hide malicious activity and maintain unauthorized access to a system

Effects of malware

Malware can :

- Steal personal or business information
- Damage or delete files
- Encrypt important data
- Slow down computers
- Monitor user activity
- Create unauthorized access

Malware Prevention :

- Keep operating systems and applications updated.
- Use reputable security/antivirus software.
- Avoid suspicious downloads and attachments.
- Don't install software from untrusted sources.
- Use strong passwords and MFA.
- Maintain regular backups.
- Scan removable storage devices before opening files.

[3] DDoS :

DDoS stand for Distributed Denial of service. It is a cyber attack in which an attacker uses many compromised devices or system to send a large amount of traffic or requests Forward target server website or network

How DDoS attack works :

Attacker → Botnet → Huge Number of Requests → Target Server → Service Disruption

A **botnet** is a collection of compromised devices controlled by an attacker. These devices can simultaneously send traffic to the target.

Example :

Imagine a website normally 1000 requests per minute. If thousands or millions of malicious requests suddenly arrives the server may become overloaded

As a result

- Website becomes very slow
- Users may revives errors
- Services may become unavailable
- Server resources can be exhausted

Common type of DDoS attacks :

Volumetric Attack – Attempts to consume the target's network bandwidth.

Protocol Attack – Exploits weaknesses in network or protocol handling.

Application-Layer Attack – Targets applications such as web servers with large numbers of requests.

[4] SQL Injection :

SQL Injection is a web application security vulnerability in which untrusted user input is improperly included in an SQL query

Allowing an attacker to manipulate the application database query

It can potentially allow unauthorized users to read modify or delete database information

How SQL injection works :

User Input → Web Application → SQL Query → Database

If the application does not properly validate or parameterize input, malicious input can alter the intended query.

Simple example :

Suppose a login system internally creates a query using user provided values

```
SELECT * FROM users
```

```
WHERE username = 'user_input'
```

```
AND password = 'password_input';
```

Possible impact :

A successful SQL injection vulnerability can potentially lead to

- Unauthorized access to database records
- Exposure of sensitive information
- Modification of database data
- Deletion of records
- Authentication bypass

[5] Brute Force Attack :

A Brute Force Attack is a cyber security in which an attacker repeatedly tries different usernames, passwords or authentication until the correct credentials are discovered

How It Works :

**Attacker → Repeated Login Attempts →
Authentication System → Correct Credentials
→ Unauthorized Access**

For example, if an account uses a weak password, an attacker may systematically try many possible passwords until one works.

Common Types

1. **Simple Brute Force** – Tries many possible password combinations.
2. **Dictionary Attack** – Uses a list of commonly used passwords and words.
3. **Credential Stuffing** – Uses leaked username/password combinations from other breaches.
4. **Password Spraying** – Tries one common password against many different accounts.

Possible Impact :

- A successful attack can result in:
- Unauthorized account access
- Theft of sensitive information
- Account takeover
- Financial or reputational damage
- Further attacks using the compromised account

Prevention :

- Organizations can protect against brute-force attacks using:
- **Strong, unique passwords**
- **Multi-factor authentication (MFA)**
- **Rate limiting**
- **Account lockout or temporary delays**
- **CAPTCHA**
- Monitoring and alerting for repeated failed logins
- Password managers

[6] Ransomware :

Ransomware is a type of malware that prevents users or organizations from accessing their files or system commonly by encrypting data and then demands a ransom in exchange for restoring access.

How Ransomware Works :

Attacker → Malware Infection → Data Encryption → Victim Loses Access → Ransom Demand

For example, ransomware may encrypt documents, images, databases and other important files so that the victim cannot open them.

Common ways Ransomware Spreads :

- Phishing emails and malicious attachments
- Malicious or compromised websites
- Vulnerable or outdated software
- Stolen or weak credentials
- Compromised remote access services

Impact of Ransomware :

A ransomware attack can cause:

- Loss of access to important files
- Business interruption
- Financial losses
- Data theft
- System downtime
- Reputational damage

How to Prevent Ransomware :

- Keep operating systems and applications updated.
- Use reputable endpoint security software.
- Avoid suspicious links and attachments.
- Use MFA and strong, unique passwords.
- Limit user and application privileges.
- Maintain **regular offline or otherwise isolated backups**.
- Test backup restoration procedures.
- Monitor networks and endpoints for suspicious activity.

3 Study Attack Vectors :

Social Engineering, Wireless Attacks, and Insider Threats.

- Social Engineering
- Wireless Attacks
- Inside Threats

What is attack vectors :

An attack vector is a method or pathway that a cyber attacker uses to gain unauthorized access to a computer, network, account or sensitive information.

[1] Social Engineering :

Social engineering is an attack technique that manipulates people into performing an action or reliving confidential information

How Social Engineering Works :

**Attacker → Builds Trust → Manipulates
Victim → Victim Takes Action →
Information/System Compromised**

Common social Engineering techniques

Phishing :

Fake emails or websites are used to trick users into revealing passwords or other sensitive information.

Spear Phishing :

A targeted phishing attack designed for a particular person or organization.

Smishing :

Phishing conducted through SMS or text messages.

Vishing :

Social engineering performed through phone or voice calls.

Pretexting :

The attacker creates a fake situation or identity to convince the victim to provide information.

Baiting :

The attacker offers something attractive, such as a free download, to encourage the victim to perform an unsafe action.

Tailgating :

An unauthorized person follows an authorized person into a restricted physical area.

Example :

An attacker may pretend to be an organization's IT employee and tell a user:

“Your account has a security problem. Please verify your login information.”

If the user provides confidential information, the attacker may use it to compromise the account.

How to Prevent Social Engineering :

- Verify unexpected requests independently.
- Never share passwords or OTPs.
- Check email addresses and URLs carefully.
- Be cautious with unexpected attachments and links.
- Use MFA.
- Provide cybersecurity awareness training.
- Follow organizational security policies.

[2] Wireless Attacks :

Wireless attacks are security attacks that target Wi-Fi networks, wireless devices, or the communication between them. Attackers may try to intercept traffic, gain unauthorized access, disrupt connectivity, or steal credentials.

Common Types of Wireless Attacks :

1. Evil Twin Attack :

An attacker creates a fake Wi-Fi access point that looks similar to a legitimate network. Users may connect to it unknowingly.

2. Deauthentication Attack

An attacker sends forged de authentication frames to disconnect devices from a Wi-Fi network. This is commonly used to demonstrate weaknesses in Wi-Fi management frames.

3. Rogue Access Point

An unauthorized access point is connected to or placed within a network, potentially allowing attackers to monitor or manipulate traffic.

4. Wi-Fi Password Attacks

Attackers may attempt to guess or crack weak Wi-Fi passwords. Strong WPA2/WPA3 passwords significantly reduce this risk.

5. Packet Sniffing

Wireless traffic is captured and analyzed to understand network communication. Unencrypted traffic can expose sensitive information.

6. Man-in-the-Middle (MITM)

An attacker positions themselves between a user and the intended network/service to potentially intercept or manipulate communication.

7. Jamming / Denial of Service

Wireless interference is used to prevent legitimate devices from communicating with an access point.

[3] Insider Threats :

Insider threats are security risks caused by people who have authorized access to an organization's systems, networks, or data but misuse that access intentionally or accidentally.

Types of Insider Threats :

1. Malicious Insider

- An employee intentionally steals, deletes, or leaks confidential information.
- Example: An employee copies customer data before leaving a company.

2. Negligent Insider

- An employee accidentally causes a security incident.
- Example: Sending a confidential document to the wrong email address.

3. Compromised Insider

- An employee's account is taken over by an external attacker.
- **Example:** An attacker steals an employee's password and uses the account to access company systems.

4. Third-Party Insider

- Contractors, vendors, or temporary workers misuse their authorized access.
- **Example:** A contractor accesses files that are not required for their work.

Common Insider Threat Activities

- Data theft
- Unauthorized file access
- Password sharing
- Installing unauthorized software
- Data leakage
- Deleting or modifying files
- Using personal USB devices
- Sharing confidential information

Warning Signs :

- Accessing unusual files
- Large amounts of data being downloaded
- Repeated failed login attempts
- Access outside normal working hours
- Unauthorized software installation
- Attempts to bypass security controls

Prevention :

1. **Least Privilege** – Give users only the permissions they need.
2. **Multi-Factor Authentication (MFA)** – Add another layer of authentication.
3. **Access Monitoring** – Monitor important system and data access.
4. **Data Loss Prevention (DLP)** – Detect and prevent unauthorized data transfers.
5. **Employee Training** – Teach employees about security policies.

6. **Strong Password Policies** – Require unique and secure passwords.

7. **Regular Auditing** – Review accounts and permissions regularly.

8. **Incident Response** – Have a procedure for investigating suspicious activity.

2 Lab Environment Setup

- Virtual box and VMware
- Kali Linux
- Metasploitable 2/DVWA
- Host only / internal network
- Lab architecture diagram

[1] VirtualBox or VMware :

VirtualBox and VMware are virtualization software used to create and run virtual machines on a physical computer.

A virtual machine is a software based computer with its own operating system storage memory and network configuration

A Virtual Machine is a software based computer running inside another physical computer

Oracle VM Virtual Box is a virtualization platform that allows users to create and run virtual machine

Common uses in cyber security

- Running Kali Linux
- Creating cybersecurity practice lab
- Testing software safely
- Learning Linux
- Creating isolated environments

VMware :

VMware provides virtualization software that can also run multiple operating systems on a single physical computer.

It is commonly used for:

- Cybersecurity labs
- Server virtualization
- Network testing
- Operating-system testing
- Enterprise environment

[2] Kali Linux as the attacker machine :

Kali Linux is a Debian based Linux distribution designed for cyber security, penetration testing, digital forensics security research and security auditing. It is maintained by Offensive Security

Kali Linux provide many security tools in one operating system. It can be installed directly on a computer or run safely inside VirtualBox and VMware as a virtual machine.

Common uses :

1. Network Security Testing – Analyze networks and identify security weaknesses.
2. Vulnerability Assessment – Find potential vulnerabilities in authorized systems.
3. Web Security Testing – Test web applications for security issues.

- 4.Password Security Testing – Assess password strength in authorized environments.
- 5.Wireless Security Testing – Audit wireless networks with permission.
- 6.Digital Forensics – Examine digital evidence.
- 7.Network Analysis – Capture and analyze network traffic.

Tool	Purpose
Nmap	Network discovery and security auditing
Wireshark	Network packet analysis
Burp Suite	Web application security testing
Metasploit Framework	Security testing and vulnerability validation
John the Ripper	Password security auditing
Nikto	Web server security assessment
Gobuster	Authorized directory/resource discover

Advantages :

- Free and open source
- Large collection of cybersecurity tools
- Useful for cybersecurity education
- Supports virtual machines
- Suitable for security testing and research
- Highly customizable Linux environment

[3] Metasploitable2 or DVWA as the target machine :

Step 1: Download Metasploitable2

Search for the official Rapid7 Metasploitable2 download page and download the VM archive.

After downloading, extract the ZIP file.

Step 2: Open VirtualBox

1. Open **Oracle VirtualBox**.
2. Click **New** or use the existing Metasploitable2 VM configuration.

3. Select the extracted Metasploitable2 virtual disk (.vmdk) when prompted.
4. Give the VM around **512 MB–1 GB RAM**.

Step 3: Configure the Network :

For a safe lab, use an **isolated Host-Only network** or another network configuration that prevents the vulnerable VM from being exposed to the Internet.

For example:

Kali Linux

192.168.56.10

|

| Host-Only Network

|



Metasploitable2

192.168.56.11

Step 4: Start Metasploitable2

Start the VM.

The default login is commonly:

Username: msfadmin

Password: msfadmin

Step 5: Find Its IP Address

Inside Metasploitable2:

ifconfig

Look for the inet address, for example:

inet addr:192.168.56.11

Step 6: Test From Kali

On Kali Linux:

ping 192.168.56.11

Then perform a basic service scan:

nmap 192.168.56.11

..

You should see services running on the intentionally vulnerable machine.

[4] Configure a private lab network (Host-Only Adapter)

Step 1 — Open Virtual Network Editor

1. Open **VMware Workstation**.
2. Go to **Edit → Virtual Network Editor**.
3. Click **Change Settings** if administrator permission is requested.
4. Find **VMnet1**.
5. Set:
 - VMnet1
 - Type: Host-only
 - Subnet IP: 192.168.56.0
 - Subnet Mask: 255.255.255.0
6. Enable **Use local DHCP service to distribute IP address to VMs** if you want VMware to automatically assign IP addresses.
7. Click **Apply → OK**.

Step 2 — Configure Kali Linux

1. Select your **Kali Linux VM**.
2. Click **Edit virtual machine settings**.
3. Select **Network Adapter**.
4. Select:
Network connection:
 - Host-only: A private network shared with the host
5. Make sure **Connected** and **Connect at power on** are enabled.
6. Click **OK**.

Step 3 — Configure Metasploitable2

Open:

Metasploitable2 → Edit virtual machine settings → Network Adapter

Select: Host-only

Make sure it uses **VMnet1**.

Step 4 — Start Kali and Check IP

Open the Kali terminal:

```
ip addr
```

You should see something similar to:

```
eth0
```

```
inet 192.168.56.10/24
```

Your actual address may be different if DHCP is enabled.

Step 5 — Check Metasploitable2

Inside Metasploitable2:

```
ifconfig
```

For example:

```
eth0
```

```
inet addr:192.168.56.11
```

Step 6 — Test the Connection

From Kali:

```
ping 192.168.56.11
```

If you get replies, your private lab network is working.

Step 7 — Test With Nmap

From Kali, scan **only your Metasploitable2 lab machine**:

```
nmap 192.168.56.11
```

You can also use:

```
nmap -sV 192.168.56.11
```

-sV attempts to identify the versions of services running on the target.

3 Linux Fundamentals

- File System Navigation (cd, ls, pwd).
- File & Directory Permissions (chmod, chown)
- Package Management (apt, dpkg)
- Networking Commands (ifconfig, ping, netstat, tracerout)

File System Navigation (cd, ls, pwd) :

There are three basic Linux commands used to navigate and view directories

1 pwd Print Working Directory

the pwd command shows the current directory
/location where you are working

syntax

pwd

Example

pwd

/home/abhi

2 Ls = list

The ls command displays the files and folders inside the current directory

Syntax

Ls

Example

ls

Documents Downloads Pictures Videos

3 cd

the cd command is used to move from one directory to another

syntax

cd directory_name

example

cd Documents

Command

Meaning

Use

pwd

Print
Working
Directory

Where am
I?

ls

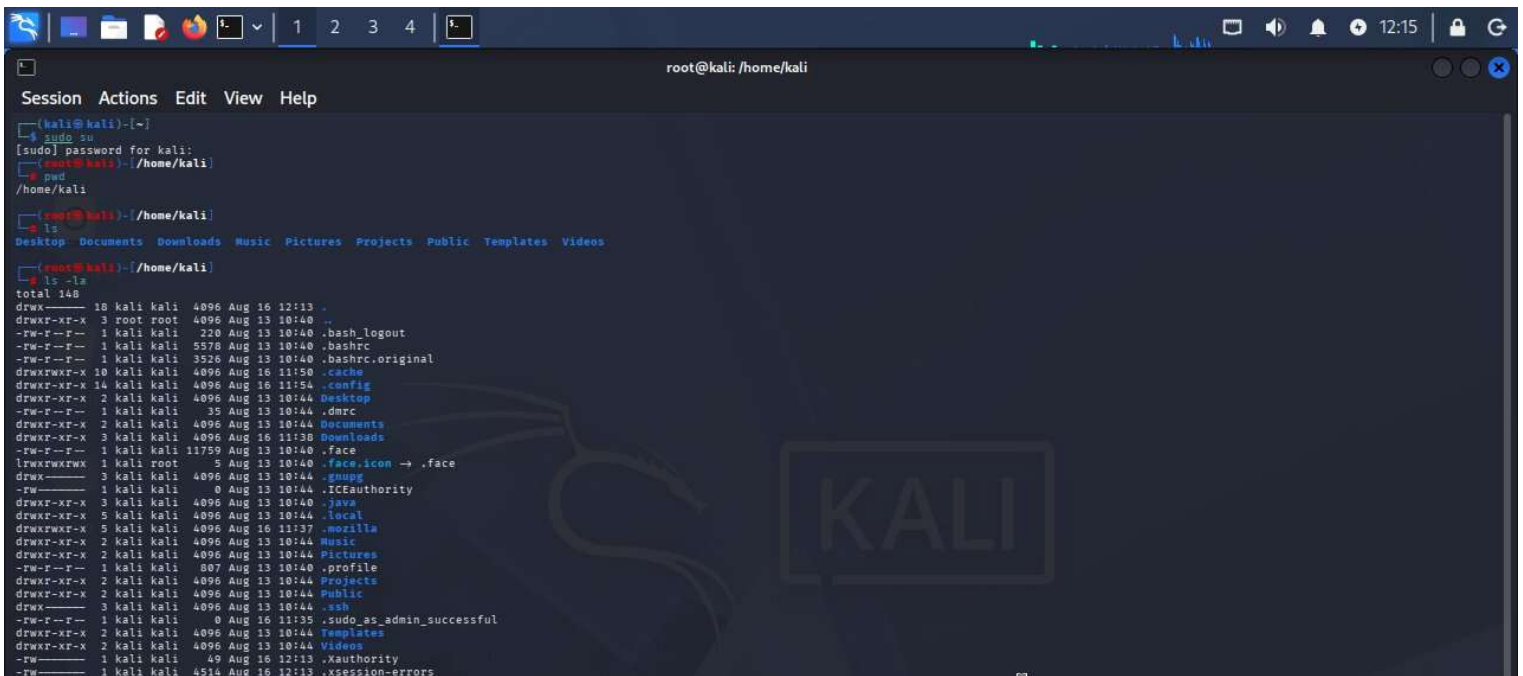
List

What is
here?

cd

Change
Directory

Move
somewhere
else



```
root@kali: /home/kali
Session Actions Edit View Help
(kali@kali)-[~]
└─$ sudo su
[sudo] password for kali:
(kali@kali)-[/home/kali]
└─$ pwd
/home/kali
(kali@kali)-[/home/kali]
└─$ ls
Desktop  Documents  Downloads  Music  Pictures  Projects  Public  Templates  Videos
(kali@kali)-[/home/kali]
└─$ ls -la
total 148
drwxr-xr-x 10 kali kali 4096 Aug 16 12:13 .
drwxr-xr-x  3 root root 4096 Aug 13 10:40 ..
-rw-r--r--  1 kali kali  220 Aug 13 10:40 .bash_logout
-rw-r--r--  1 kali kali 5578 Aug 13 10:40 .bashrc
-rw-r--r--  1 kali kali 3526 Aug 13 10:40 .bashrc.original
drwxr-xr-x 10 kali kali 4096 Aug 16 11:50 .cache
drwxr-xr-x 14 kali kali 4096 Aug 16 11:54 .config
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 Desktop
-rw-r--r--  1 kali kali  35 Aug 13 10:44 .dmrc
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 Documents
drwxr-xr-x  3 kali kali 4096 Aug 16 11:30 Downloads
-rw-r--r--  1 kali kali 11759 Aug 13 10:40 .face
-rw-r--r--  1 kali root  5 Aug 13 10:40 .face.icon -> .face
drwxr-xr-x  3 kali kali 4096 Aug 13 10:44 .gnupg
-rw-r--r--  1 kali kali  0 Aug 13 10:44 .ICEauthority
drwxr-xr-x  3 kali kali 4096 Aug 13 10:40 .java
drwxr-xr-x  5 kali kali 4096 Aug 13 10:44 .local
drwxr-xr-x  5 kali kali 4096 Aug 16 11:37 .mozilla
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 .Music
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 .Pictures
-rw-r--r--  1 kali kali 807 Aug 13 10:40 .profile
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 .Projects
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 .Public
drwxr-xr-x  3 kali kali 4096 Aug 13 10:44 .ssh
-rw-r--r--  1 kali kali  0 Aug 16 11:35 .sudo_as_admin_successful
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 .Templates
drwxr-xr-x  2 kali kali 4096 Aug 13 10:44 .Videos
-rw-r--r--  1 kali kali  49 Aug 16 12:13 .Xauthority
-rw-r--r--  1 kali kali 4514 Aug 16 12:13 .Xsession-errors
```

file & Directory Permissions: chmod, chow

1 chmod – change Mode

Chmod is used to change the permission of file or directory

Symbol	Permission	Meaning
r	Read	View/read the file
w	Write	Modify the file
x	Execute	Run the file

Syntax

chmod permissions filename

Example

```
chmod 755 script.sh
```

Here

7 = Read + Write + Execute

5 = Read + Execute

5 = Read + Execute

2 shown – change owner

Shown is used to change the owner of file or directory

Syntax

```
chown username filename
```

Example

```
sudo chown abhi report.txt
```

change Owner and group

..

```
sudo chown abhi:developers report.txt
```

this change

owner = windows

group = developers


```
root@kali: /home/kali/linux_practical

Session Actions Edit View Help

root@kali:~/home/kali# mkdir linux_practical
root@kali:~/home/kali# cd linux_practical
root@kali:~/home/kali/linux_practical# pwd
/home/kali/linux_practical
root@kali:~/home/kali/linux_practical# nano test.txt
root@kali:~/home/kali/linux_practical# ls -l test.txt
-rw-r--r-- 1 root root 147 Aug 16 12:19 test.txt
root@kali:~/home/kali/linux_practical# chmod 755 test.txt
root@kali:~/home/kali/linux_practical# ls -l test.txt
-rwxr-xr-x 1 root root 147 Aug 16 12:19 test.txt
root@kali:~/home/kali/linux_practical# chmod 644 test.txt
root@kali:~/home/kali/linux_practical# ls -l test.txt
-rw-r--r-- 1 root root 147 Aug 16 12:19 test.txt
root@kali:~/home/kali/linux_practical# whoami
root
root@kali:~/home/kali/linux_practical# sudo chown $USER:$USER test.txt
root@kali:~/home/kali/linux_practical# ls -l test.txt
-rw-r--r-- 1 root root 147 Aug 16 12:19 test.txt
root@kali:~/home/kali/linux_practical# mkdir test_directory
root@kali:~/home/kali/linux_practical# ls -ld test_directory
drwxr-xr-x 2 root root 4096 Aug 16 12:22 test_directory
root@kali:~/home/kali/linux_practical# chmod 644 test_directory
root@kali:~/home/kali/linux_practical# ls -ld test_directory
drw-r--r-- 2 root root 4096 Aug 16 12:22 test_directory
root@kali:~/home/kali/linux_practical# chmod 755 test_directory
root@kali:~/home/kali/linux_practical# ls -ld test_directory
drwxr-xr-x 2 root root 4096 Aug 16 12:22 test_directory
```

[3] PACKAGE MANAGEMENT (apk,dpkg) :

Package management in Linux is used to install update remove and manage software packages

In Debian based distributes such as ubuntu and kali Linux two important package management commands are apt and dpkg

1 apt

Apt is high level package management tool. It is used can automatically handle software dependencies

Install a package

```
sudo apt install package_name
```

example

```
sudo apt install wireshark
```

..

update package information

```
sudo apt update
```

update installed packages

```
sudo apt upgrade
```

remove a package

```
sudo apt remove package_name
```

example

```
sudo apt remove wireshark
```

search for a package

```
apt search package_name
```

2 dpkg

Dpkg is a low level package management tool used to work directly with .deb package files

Install a .deb file

```
sudo dpkg -i package.deb
```

check installed packages

```
dpkg -l
```

find information about a package

```
dpkg -s package_name
```

remove a package

```
sudo dpkg -r package_name
```

```
root@kali: /home/kali
Session Actions Edit View Help

root@kali: /home/kali
# sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [36.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [115 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [265 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.4 MB in 56s (1,373 kB/s)
1056 packages can be upgraded. Run 'apt list --upgradable' to see them.

root@kali: /home/kali
# apt search nmap
Error: Invalid operation serach

root@kali: /home/kali
# apt search nmap
atac/kali-rolling 0-20150903+r2013-10+b3 amd64
genome assembly-to-assembly comparison

brutespray/kali-rolling 2.2.2-0kali1 amd64
Bruteforcing from various scanner output

dscan/kali-rolling 0.1.5-0kali1 all
wrapper around nmap

dscan-doc/kali-rolling 0.1.5-0kali1 all
wrapper around nmap (documentation)

eyewitness/kali-rolling,now 20230525-1+git20230720-0kali4 amd64 [installed,automatic]
Rapid web application triage tool

fierce/kali-rolling,now 1.6.0-2 all [installed,automatic]
Domain DNS scanner

golang-github-lair-framework-go-nmap-dev/kali-rolling 0.0-g+git20191202.3507e0b-1.1 all
nmap XML parsing library for Go (library)

golang-github-malfunkt-sprange-dev/kali-rolling 0.9.0-5 all
IPv4 address parser for the nmap format

incus-extra/kali-rolling 7.0.1-2 amd64
Modern container and virtual machine manager - extra tools

ivre/kali-rolling 0.9.21-0kali1 all
network recon framework IVRE or DRUNK

ivre-doc/kali-rolling 0.9.21-0kali1 all
Documentation for IVRE package

libfile-map-perl/kali-rolling 0.71-1-b1 amd64
Perl module providing simple and safe memory mapping

libnmap-parser-perl/kali-rolling 1.37-2 all
module to parse nmap scan results with perl

librust-cpp-synmap-dev/kali-rolling 0.3.0-1+b3 amd64
Sourcecode and full crate parsing support for "cpp_syn" - Rust source code

lwm-tools-nmap/kali-rolling 0.0-g+git20250612.36818ca-0kali1 all
plugin for LWM tool hat provides Nmap network scanning capabilities
```

```
root@kali: /home/kali
Session Actions Edit View Help

root@kali: /home/kali
# dpkg -i | grep nmap
dpkg: error: --install needs at least one package archive file argument

Type dpkg --help for help about installing and deinstalling packages [*];
Use 'apt' or 'aptitude' for user-friendly package management;
Type dpkg --get-selections for a list of packages;
Type dpkg --get-architecture for a list of architectures;
Type dpkg --force-help for a list of forcing options;
Type dpkg --get-architecture for a list of architectures;
Type dpkg --get-architecture for a list of architectures;
Type dpkg --get-architecture for a list of architectures;

Options marked [*] produce a lot of output - pipe it through 'less' or 'more' !

root@kali: /home/kali
# sudo apt install nmap
nmap is already the newest version (7.99+dfsg-1kali1).
nmap set to manually installed.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1056

root@kali: /home/kali
# dpkg -s nmap
Package: nmap
Status: install ok installed
Priority: optional
Section: non-free/net
Installed-Size: 4823
Maintainer: Kali Developers <dev@kali.org>
Architecture: amd64
Version: 7.99+dfsg-1kali1
Depends: nmap-common (= 7.99+dfsg-1kali1), libc6 (>= 2.38), libgcc-s1 (>= 3.0), liblinear4 (>= 2.01+dfsg), liblua5.4-0 (>= 5.4.6), libpcap0.8t64 (>= 1.5.1), libpcre2-8-0 (>= 10.3
2), libssh2-1t64 (>= 1.11.0), libssl3t64 (>= 3.0.0), libstdc++6 (>= 5.2), zlib1g (>= 1:1.1.4)
Suggests: ncft, nmap, zenmap
Description: The Network Mapper
Nmap is a utility for network exploration or security auditing. It
supports ping scanning (determine which hosts are up), many port
scanning techniques, version detection (determine service protocols
and application versions listening behind ports), and TCP/IP
fingerprinting (remote host OS or device identification). Nmap also
offers flexible target and port specification, decoy/stealth scanning,
sunRPC scanning, and more. Most Unix and Windows platforms are
supported in both GUI and commandline modes. Several popular handheld
devices are also supported, including the Sharp Zaurus and the iPAQ.
Homepage: https://nmap.org/
Original-Maintainer: Debian Security Tools <team+pkg-security@tracker.debian.org>

root@kali: /home/kali
# dpkg -l nmap
Desired=Unknown/Install/Remove/Purge/Hold
Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
Name Version Architecture Description
```

```
root@kali: /home/kali
Session Actions Edit View Help
ii nmap 7.99+dfsg-1kali1 amd64 The Network Mapper

(root@kali) ~/home/kali
$ sudo apt remove nmap
The following packages were automatically installed and are no longer required:
binutils-mingw-w64-base httpx-toolkit ndiff python3-dunamai python3-poetry-core python3-tld
binutils-mingw-w64-1686 imagemagick nmap-common python3-fuzzywuzzy python3-poetry-dynamic-versioning python3-tomlkit
binutils-mingw-w64-x86-64 imagemagick-7.q16 nuclei python3-github python3-pyexploitdb python3-unicrypto
bloodhound.py krb5-multidev oracle-instantclient-basic python3-github python3-pyfiglet python3-wapiti-swagger
comerr-dev libksrppc4t64 python3-aardwolf python3-httpx-mtlm python3-pylink3 python3-winacl
dnsmasq libkadm5clnt-mit12 python3-aesedb python3-jespney python3-pynfsclient python3-winacl
dsniff libkadm5srv-mit12 python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
ettercap-common libkdb5-10t64 python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
ettercap-graphical libkdb5-dev python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
eyewitness liblinea4 python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
feroxbuster liblua5.1-2 python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
figlet liblua5.1-common python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
finger libnids1.2t64 python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
gcc-mingw-w64-base libnids1.2t64 python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
gcc-mingw-w64-1686-win32 medusa python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
gcc-mingw-w64-1686-win32-runtime mingw-w64-common python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
gcc-mingw-w64-x86-64-win32 mingw-w64-1686-dev python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
gcc-mingw-w64-x86-64-win32-runtime mingw-w64-x86-64-dev python3-aioconsole python3-krb5 python3-pypsp python3-xmldict
Use 'sudo apt autoremove' to remove them.

REMOVING:
commix kali-linux-default kali-linux-headless kali-tools-top10 legion metasploit-framework msfpc netexec nmap python3-libnmap set unicorn-magic zenmap

Summary:
Upgrading: 0, Installing: 0, Removing: 13, Not Upgrading: 1051
Freed space: 660 MB

Continue? [Y/n] y
(Reading database.. 425968 files and directories currently installed.)
Removing kali-linux-default (2026.2.12)..
Removing kali-linux-headless (2026.2.12)..
Removing commix (4.1-0kali1)..
Removing kali-tools-top10 (2026.2.12)..
Removing legion (0.7.0-0kali2)..
Removing set (8.0.3+git20241021-0kali1)..
Removing unicorn-magic (3.12-0kali3)..
Removing msfpc (1.4.5-0kali3)..
Removing netexec (1.5.1-0kali2)..
Removing zenmap (7.99+dfsg-1kali1)..
Removing python3-libnmap (0.7.3-1)..
Removing metasploit-framework (6.4.135-0kali1)..
Removing nmap (7.99+dfsg-1kali1)..
Processing triggers for desktop-file-utils (0.28-1)..
Processing triggers for hicolor-icon-theme (0.18-2)..
Processing triggers for libc-bin (2.42-16)..
Processing triggers for man-db (2.13.1-1)..
Processing triggers for wordlists (2026.2.0)..
Processing triggers for kali-menu (2026.2.6)..
```

4 networking commands

Linux provides several commands to check network configuration test connectivity and troubleshoot network problems. Four important commands are ifconfig, ping, netstat and traceroute

1 ifconfig – interface configuration

Ifconfig is used to view and configure network interfaces

Syntax

Ifconfig

It displays information such as:

- IP address
- MAC address
- Network interface status
- Received/transmitted packets

Example

Ifconfig

2 ping – test connectivity

Ping checks wheather another host is reachable over a network

Syntax

ping destination

Example

ping google.com

Uses

- Check Internet connectivity
- Check whether a server is reachable
- Measure approximate response time
- Identify packet loss

Stop the command with : Ctrl + c

3 netstat – network statistics

Netstat displays information about network connections listening ports routing information and network statistics

Example

```
netstat -an
```

common options

```
netstat -tuln
```

4 traceroute – trace network path

traceroute shows the path packets take from your computer to a destination

syntax

```
traceroute destination
```

example

```
traceroute google.com
```

It displays the intermediate **routers/hops** between your computer and the destination.

Uses

- Find where network delays occur
- Identify routing problems
- Troubleshoot connectivity issues
- Understand the path to a remote se

```
root@kali: /home/kali
Session Actions Edit View Help

root@kali: /home/kali
# ifconfig
docker0: flags=4096<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.0.255
    ether 02:42:c0:00:00:00 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=1601<UP,BROADCAST,MULTICAST> mtu 1500
    inet 192.168.1.77 netmask 255.255.255.0 broadcast 192.168.29.255
    inet6 2405:201:200:801::201:29ff:fe09:7e4d prefixlen 64 scopeid 0<global>
    inet6 fe80::20c:29ff:fe09:7e4d prefixlen 64 scopeid 0<link>
    inet6 2405:201:200:801::201:29ff:fe09:7e4d prefixlen 64 scopeid 0<global>
    ether 08:00:20:c0:00:00 txqueuelen 1000 (Ethernet)
    RX packets 57769 bytes 8129648 (77.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 27918 bytes 2239363 (2.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@kali: /home/kali
# ping -c 4 google.com
PING google.com (1406:0000:0000:0000::1) 32 data bytes
64 bytes from icmp-in-f110.1e100.net (2404:0000:0000:0000::1): icmp_seq=1 ttl=189 time=23.1 ms
64 bytes from icmp-in-f110.1e100.net (2404:0000:0000:0000::1): icmp_seq=2 ttl=189 time=33.1 ms
64 bytes from icmp-in-f110.1e100.net (2404:0000:0000:0000::1): icmp_seq=3 ttl=189 time=32.8 ms
64 bytes from icmp-in-f110.1e100.net (2404:0000:0000:0000::1): icmp_seq=4 ttl=189 time=33.8 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 300ms
rtt min/avg/max/mdev = 32.825/36.312/23.112/62.447 ms

root@kali: /home/kali
# netstat -tlnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address   Foreign Address  State           PID/Program name
tcp        0      0 127.0.0.1:55577 0.0.0.0:*         LISTEN          0/*
udp        0      0 fe80::20c:29ff:fe09:7e4d:546 :::*               0/*

root@kali: /home/kali
# ss -tlnp
Netid  State     Recv-Q      Send-Q           Local Address:Port      Peer Address:Port
tcp    LISTEN     0            4096             127.0.0.1:55577        0.0.0.0:*

root@kali: /home/kali
# traceroute google.com
traceroute to google.com (2404:0000:0000:0000::1): 30 hops max, 60 byte packets
 0  * * *
 1  2405:201:200:801::201 [2405:201:200:801::201:29ff:fe09:7e4d] 12.409 ms 43.545 ms 43.173 ms
 2  * * *
 3  2405:201:200:801::201 [2405:201:200:801::201:29ff:fe09:7e4d] 26.409 ms 59.254 ms 59.247 ms
 4  2405:201:200:801::201 [2405:201:200:801::201:29ff:fe09:7e4d] 61.494 ms 61.419 ms 61.360 ms
 5  2405:201:200:801::201 [2405:201:200:801::201:29ff:fe09:7e4d] 65.252 ms 69.906 ms 72.136 ms
 6  * * *
 7  2405:201:200:801::201 [2405:201:200:801::201:29ff:fe09:7e4d] 44.858 ms 48.331 ms 2405:201:200:801::201 [2405:201:200:801::201:29ff:fe09:7e4d] 48.219 ms
 8  * * *
 9  2001:4000:1::1038 [2001:4000:1::1038] 67.746 ms * 2001:4000:1::1038 [2001:4000:1::1038] 77.246 ms
10  2001:4000:1::1038 [2001:4000:1::1038] 77.135 ms * 2001:4000:1::1038 [2001:4000:1::1038] 26.508 ms
11  2001:4000:1::1038 [2001:4000:1::1038] 29.218 ms 2404:0000:0000:0000::1 [2404:0000:0000:0000::1] 38.851 ms 2001:4000:1::1038 [2001:4000:1::1038] 32.802 ms
12  2001:4000:1::1038 [2001:4000:1::1038] 36.225 ms 2001:4000:1::1038 [2001:4000:1::1038] 20.373 ms 2001:4000:1::1038 [2001:4000:1::1038] 38.500 ms
13  * 2001:4000:1::1038 [2001:4000:1::1038] 33.767 ms 36.473 ms
14  2001:4000:1::1038 [2001:4000:1::1038] 36.822 ms * 2001:4000:1::1038 [2001:4000:1::1038] 38.850 ms
15  * 2001:4000:1::1038 [2001:4000:1::1038] 32.049 ms *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  icmp-in-f110.1e100.net (2404:0000:0000:0000::1) 38.808 ms * *
```

Networking basics

- OSI Model Layers & Functions
- TCP/IP Protocol Suite.
- DNS & HTTP/HTTPS Deep Dive.
- IP Addressing, Subnetting, and NAT

1 OSI Model Layers & Functions

The OSI (Open Systems Interconnection) Model is a reference model that explains how data is communicated between two devices over a network. It has 7 layers.

OSI Model Layers

7. Application

6. Presentation

5. Session

4. Transport

3. Network

2. Data Link

1. Physical

1. Physical Layer

The **Physical Layer** is responsible for transmitting **raw bits (0s and 1s)** through the physical medium.

Functions:

- Transmission of bits
- Defines cables, connectors and signals
- Specifies data transmission speed
- Handles physical network devices

Examples: Ethernet cable, fiber-optic cable, hubs.

2. Data Link Layer

The **Data Link Layer** provides reliable communication between directly connected devices.

Functions:

- Creates **frames**
- Uses MAC addresses
- Error detection
- Controls access to the transmission medium

Examples: Ethernet, switches, MAC address

3. Network Layer

The **Network Layer** is responsible for **logical addressing and routing** packets between networks.

Functions:

- IP addressing
- Routing
- Packet forwarding
- Path selection

Examples: IP, routers, IPv4, IPv6.

4. Transport Layer

The **Transport Layer** provides **end-to-end communication** between applications.

Functions:

- Segmentation and reassembly
- Flow control
- Error recovery
- Reliable or unreliable delivery
- Port addressing

Examples: TCP and UDP.

5. Session Layer

The **Session Layer** establishes, manages, and terminates communication sessions between applications.

Functions:

- Session establishment
- Session management
- Session termination
- Synchronization

Example: Managing a communication session between applications.

6. Presentation Layer

The **Presentation Layer** handles the **format and representation of data**.

Functions:

- Data translation
- Encryption and decryption
- Data compression
- Data formatting

Examples: JPEG, PNG, encryption formats, character encoding

7. Application Layer

The **Application Layer** provides network services directly to **user applications**.

Functions:

- Provides network services to applications
- Email communication
- Web access
- File transfer
- Name resolution

Examples: HTTP/HTTPS, FTP, DNS, SMTP.

2 TCP/IP Protocol Suite :

The TCP/IP Protocol Suite is a collection of communication protocols used to connect devices and exchange data over networks, including the Internet. TCP/IP stands for Transmission Control Protocol/Internet Protocol. It is commonly represented using 4 layers.

1. Network Access Layer

This is the lowest layer. It is responsible for transferring data between devices over the physical network.

Functions:

- Sends data over the network medium
- Handles frames and MAC addresses
- Provides access to the physical network

Examples: Ethernet, Wi-Fi, ARP.

2. Internet Layer

The Internet Layer is responsible for **logical addressing and routing packets** between networks.

Functions:

- Provides IP addressing
- Determines the path for packets
- Transfers packets between networks

Protocols: IPv4, IPv6, ICMP.

Device: Router.

3. Transport Layer

The Transport Layer provides **end-to-end communication** between applications.

It mainly uses two protocols:

TCP

- Connection-oriented
- Reliable delivery
- Error checking
- Ensures packets arrive in the correct order

UDP

- Connectionless
- Faster than TCP
- No guarantee of delivery
- Suitable for real-time applications

Examples: TCP, UDP.

4. Application Layer

The Application Layer provides network services directly to applications and users.

Common Protocols

Protocol	Purpose
HTTP/HTTPS	Web browsing
FTP	File transfer
SMTP	Sending email
DNS	Domain-name resolution
SSH	Secure remote access
DHCP	Automatic IP configuration

[3] DNS & HTTP/HTTPS Deep Dive :

1 DNS – Domain Name system

DNS (Domain Name system) is a system that converts domain names into IP address so computers can locate servers on a network

For example

www.example.com



DNS



93.184.216.34

Humans remember domain names easily, while computers communicate using IP addresses.

Important DNS Components

1. DNS Resolver

Receives the DNS request from the client and searches for the required IP address.

2. Root Server

Directs the resolver toward the appropriate Top-Level Domain (TLD) server.

3. TLD Server

Handles domains such as .com, .org, .net, etc.

4. Authoritative DNS Server

Contains the actual DNS records for a domain.

Common DNS records

Record	Purpose
A	Domain → IPv4 address
AAAA	Domain → IPv6 address
CNAME	Alias for another domain
MX	Mail server information
NS	Name server information
TXT	Text/verification information

[2] HTTP – Hypertext Transfer Protocol :

HTTP is an application layer protocol used to transfer web resources between client and web server

..

Example

Browser

|

| HTTP Request

↓

Web Server

|

| HTTP Response

↓

Browser

HTTP request

A browser may send a request such as

GET /index.html HTTP/1.1

Host: example.com

HTTP response

HTTP/1.1 200 OK

Common HTTP Methods

Method	Purpose
GET	Retrieve data
POST	Send/create data
PUT	Update/replace data
PATCH	Partially update data
DELETE	Delete data

Common HTTP status codes

Code	Meaning
200	OK / successful
201	Created
301	Permanently moved
400	Bad request
401	Unauthorized
403	Forbidden

..

Code Meaning

404 Not found

500 Internal server error

[3] HTTPS – secure HTTP :

HTTPS is a HTTP protected using TCL Transport layer Security

HTTP

+

TLS encryption

↓

HTTPS

[4] IP Addressing, Subnetting, and NAT :

These are important concepts in computer networking. IP addressing identifies devices, subnetting divides networks into smaller networks, and NAT translates private and public IP addresses.

1 IP Address

An IP address (Internet Protocol address) is a logical address assigned to a device on a network. It allows devices to identify and communicate with each other.

IPv4 Address

An IPv4 address contains **32 bits**, divided into four octets.

Example: 192.168.1.10

Each octet can have a value from 0 to 255.

The exact division between network and host depends on the subnet mask/prefix length.

Types of IP Addresses

1. Public IP

Used for communication over the Internet.

2. Private IP

Used inside local networks.

Common private IPv4 ranges:

10.0.0.0/8

172.16.0.0/12

192.168.0.0/16

3. Static IP

Manually configured or reserved so it generally remains the same.

4. Dynamic IP

Usually assigned automatically using DHCP.

2. Subnetting

Subnetting is the process of dividing one IP network into **smaller logical networks called subnets**.

For example:

192.168.1.0/24



Divide into smaller networks



192.168.1.0/26

192.168.1.64/26

192.168.1.128/26

192.168.1.192/26

Subnet Mask

A subnet mask determines which part of an IP address represents the network and which part represents the host.

Example: IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

CIDR: /24

Why Subnetting Is Used

- Efficient use of IP addresses
- Divides large networks into smaller networks
- Reduces unnecessary broadcast traffic
- Improves network organization
- Can help with network security and isolation

Simple Example

For:

192.168.1.0/24

there are 256 total IPv4 addresses.

Traditionally, in a normal subnet, 254 are usable host addresses, because the first address is the network address and the last is the broadcast address.

3. NAT — Network Address Translation

NAT translates IP addresses between different networks, commonly allowing devices with private IP addresses to communicate with the Internet using a public IP address.

The router translates the internal private addresses to a public address when traffic goes to the Internet.

Types of NAT

1. Static NAT

Maps one private IP address to one public IP address.

2. Dynamic NAT

Maps private addresses to addresses from a pool of public addresses.

3. PAT (Port Address Translation)

Allows many private devices to share one public IP address by using different port numbers. This is very common in home networks.

Advantages of NAT

- Conserves public IPv4 addresses
- Allows private networks to access the Internet
- Hides internal private addresses from direct Internet addressing
- Makes it possible for many devices to share a public IPv4 address

5 Cryptography Basics

- Symmetric vs Asymmetric Encryption.
- Hashing (MD5, SHA256).
- Digital Certificates & SSL/TLS.
- Hands-on: Encrypt and Decrypt messages using OpenSSL.

1 Symmetric vs Asymmetric Encryption

Feature	Symmetric Encryption	Asymmetric Encryption
Keys used	One shared key for encryption and decryption	Two keys: a public key and a private key
Speed	Very fast	Slower due to complex mathematics
Security challenge	Securely sharing the secret key	Public key can be shared openly; private key remains secret
Typical use	Encrypting large amounts of data	Key exchange, digital signatures, authentication
Examples	AES, ChaCha20, DES (legacy)	RSA, ECC (Elliptic Curve Cryptography)

Symmetric Encryption

- The **same secret key** encrypts and decrypts the data.
- Both the sender and receiver must know this key beforehand.

Example:

- Alice and Bob both have the secret key **K**.
- Alice encrypts a message using **K**.
- Bob decrypts it using the same **K**.

Advantages

- Very fast and efficient.
- Suitable for encrypting files, databases, VPN traffic, and disk storage.

Disadvantages

- Securely distributing the secret key can be difficult.
- Doesn't inherently provide non-repudiation (proof of who sent a message).

Asymmetric Encryption

- Uses a **key pair**:
 - **Public key**: Shared with anyone.
 - **Private key**: Kept secret by the owner.

Example:

- Bob publishes his public key.
- Alice encrypts a message with Bob's public key.
- Only Bob can decrypt it using his private key.

Advantages

- No need to share a secret key beforehand.
- Enables digital signatures and authentication.

Disadvantages

- Much slower than symmetric encryption.
- Not efficient for encrypting large amounts of data.

2 Hashing (MD5, SHA256)

Hashing is a cryptographic technique that converts data of any size into a fixed-length string of characters, called a hash value or digest.

Unlike encryption, hashing is generally **one-way**—the original data should not be directly recoverable from the hash.

1. MD5 (Message-Digest Algorithm 5)

- Produces a **128-bit (32 hexadecimal character)** hash.
- It was widely used for file integrity checks and older password systems.
- MD5 is no longer considered secure because practical collision attacks exist.

- It should not be used for password storage or security-sensitive applications.

Example:

Hello → 8b1a9953c4611296a827abf8c47804d7

2. SHA-256

- SHA-256 belongs to the **SHA-2 family**.
- Produces a **256-bit (64 hexadecimal character)** hash.
- It is significantly stronger than MD5 and is widely used for integrity verification, certificates, digital signatures, and many security applications.
- For passwords, however, **SHA-256 alone is not recommended**; password-specific algorithms such as Argon2id, bcrypt, or scrypt are preferred.

Example:

Hello →

185f8db32271fe25f561a6fc938b2e264306ec304
eda518007d1764826381969

3 Digital Certificates & SSL/TLS

1. Digital Certificate

A digital certificate is an electronic document used to prove the identity of a website, server, organization, or person. It helps establish trust between two parties over a network.

Digital certificates are commonly based on X.509 standards and are issued by a trusted **Certificate Authority (CA)**.

Main Information in a Digital Certificate

- **Subject:** Identity of the certificate holder.
- **Public Key:** Used for cryptographic operations.
- **Issuer:** Certificate Authority that issued the certificate.
- **Validity Period:** Start and expiration dates.
- **Serial Number:** Unique certificate identifier.
- **Digital Signature:** CA's signature used to verify the certificate's authenticity.

- **Domain Name:** The website/domain covered by the certificate.

How a Digital Certificate Works

Website



Requests Certificate



Certificate Authority (CA)



Issues Digital Certificate



Browser Receives Certificate



Verifies CA Signature & Domain



Secure Connection Established

2. SSL/TLS

SSL (Secure Sockets Layer) and **TLS (Transport Layer Security)** are cryptographic protocols designed to secure communication over a network.

TLS is the modern successor to SSL. SSL versions are obsolete and should not be used.

TLS provides three major security properties:

1. **Confidentiality** – prevents unauthorized people from reading data.
2. **Integrity** – detects unauthorized modification of data.
3. **Authentication** – helps verify the identity of the server.

Example: HTTPS

When you visit:

`https://example.com`

HTTPS uses **HTTP over TLS** to protect communication between your browser and the website.

[3] Hands-on:

Encrypt and Decrypt messages using OpenSSL.

OpenSSL is a command-line toolkit used for cryptography, including encryption, decryption, hashing, certificates, and TLS.

For a beginner hands-on exercise, we can use AES-256 symmetric encryption

1. Check OpenSSL Installation

Open a terminal and run:

```
openssl version
```

Example output:

```
OpenSSL 3.x.x
```

2. Create a Message File

Create a file named message.txt:

This is a confidential message.

You can create it from the terminal:

echo "This is a confidential message." >
message.txt

Check the file:

cat message.txt

3. Encrypt the Message

Use AES-256-CBC encryption:

openssl enc -aes-256-cbc -salt -pbkdf2 -in
message.txt -out message.enc

OpenSSL will ask you for a password.

Enter password:

Verifying - Enter password:

The encrypted file is:

message.enc

You can check it:

ls

The encrypted file will contain ciphertext rather than the readable message.

[4] Decrypt the Message

Run:

```
openssl enc -d -aes-256-cbc -pbkdf2 -in  
message.enc -out decrypted.txt
```

Enter the **same password** used during encryption.

Then view the decrypted message:

```
cat decrypted.txt
```

Output:

This is a confidential message.

6 Tool Familiarization

- Wireshark (packet capture).
- Nmap (network scanning).
- Burp Suite (web proxy).
- Netcat (network debugging).

1 Wireshark (packet capture)

Wireshark is a free and open source network packet analyzer. It captures network packets and allows users to examine how data travels between computers, servers and other network devices

It is commonly used for

- Network troubleshooting
- Cybersecurity analysis
- Protocol learning
- Detecting unusual network activity
- Network performance analysis

What is packet capture

Packet capture is the process of recording network packets transmitted through a network interface.

3. Steps to Capture Packets

Step 1: Open Wireshark.

Step 2: Select your active network interface:

- Wi-Fi
- Ethernet

Step 3: Start the capture.

Step 4: Generate normal network traffic, such as opening a website or performing an authorized ping.

Step 5: Stop the capture.

Step 6: Select individual packets to examine their details.

4. Important Wireshark Fields

Field	Meaning
No.	Packet number
Time	Time packet was captured
Source	Sender's address
Destination	Receiver's address
Protocol	Protocol used
Length	Packet size
Info	Additional packet information

5. Useful Display Filters

Wireshark can filter captured traffic using display filters.

ICMP packets:

icmp

DNS packets:

dns

TCP packets:

tcp

UDP packets:

udp

HTTP packets:

http

Specific IP address:

ip.addr == 192.168.1.10

6. Example: Capturing ICMP Packets

Open a terminal and run a normal ping:

ping 8.8.8.8

Then use the following Wireshark filter:

icmp

You may see:

Your Computer

|

| ICMP Echo Request

..



Destination



| ICMP Echo Reply



Your Computer

This demonstrates how a device sends a request and receives a response.

7. Packet Analysis

When you select a packet, Wireshark displays protocol layers such as:

Frame



Ethernet



IP



TCP / UDP / ICMP

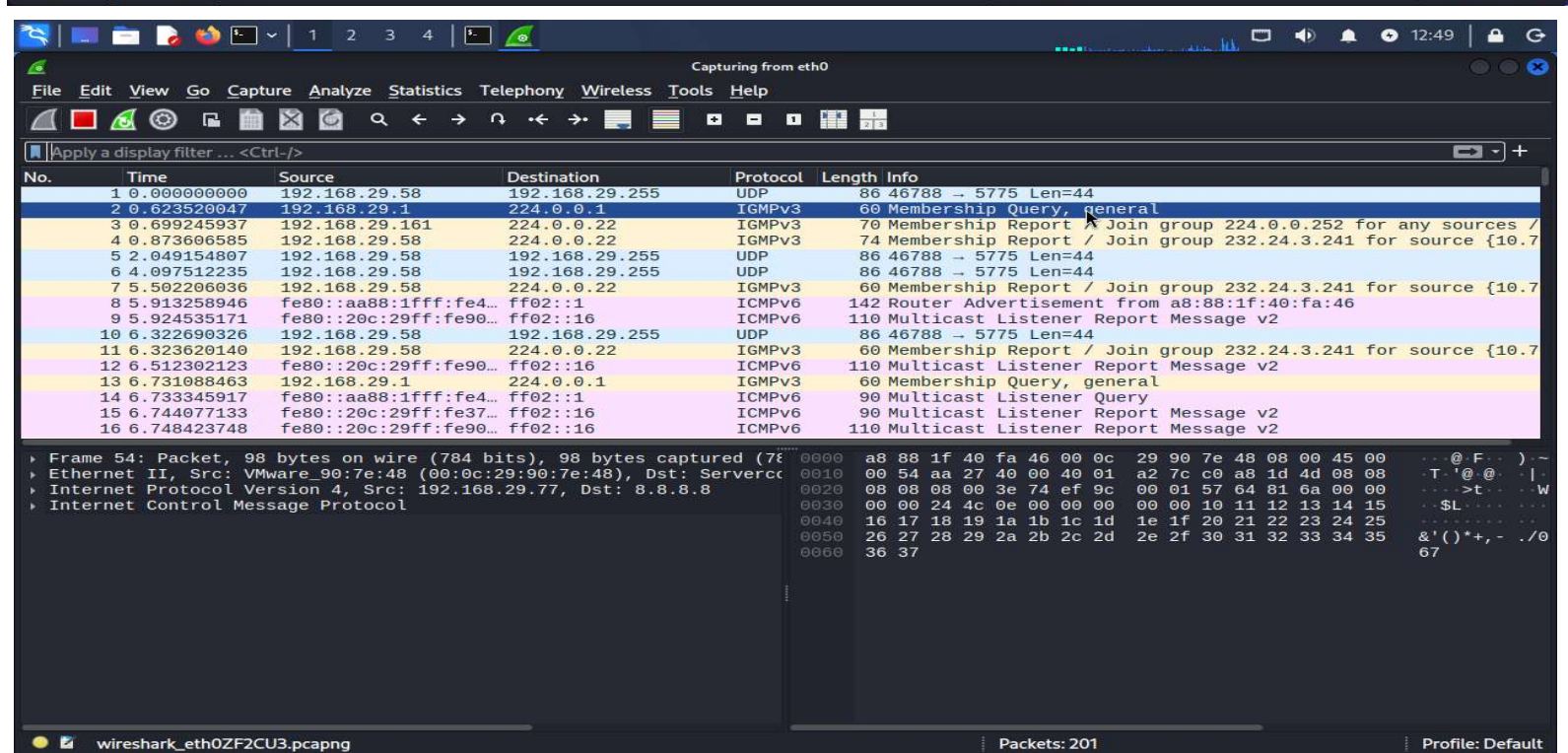
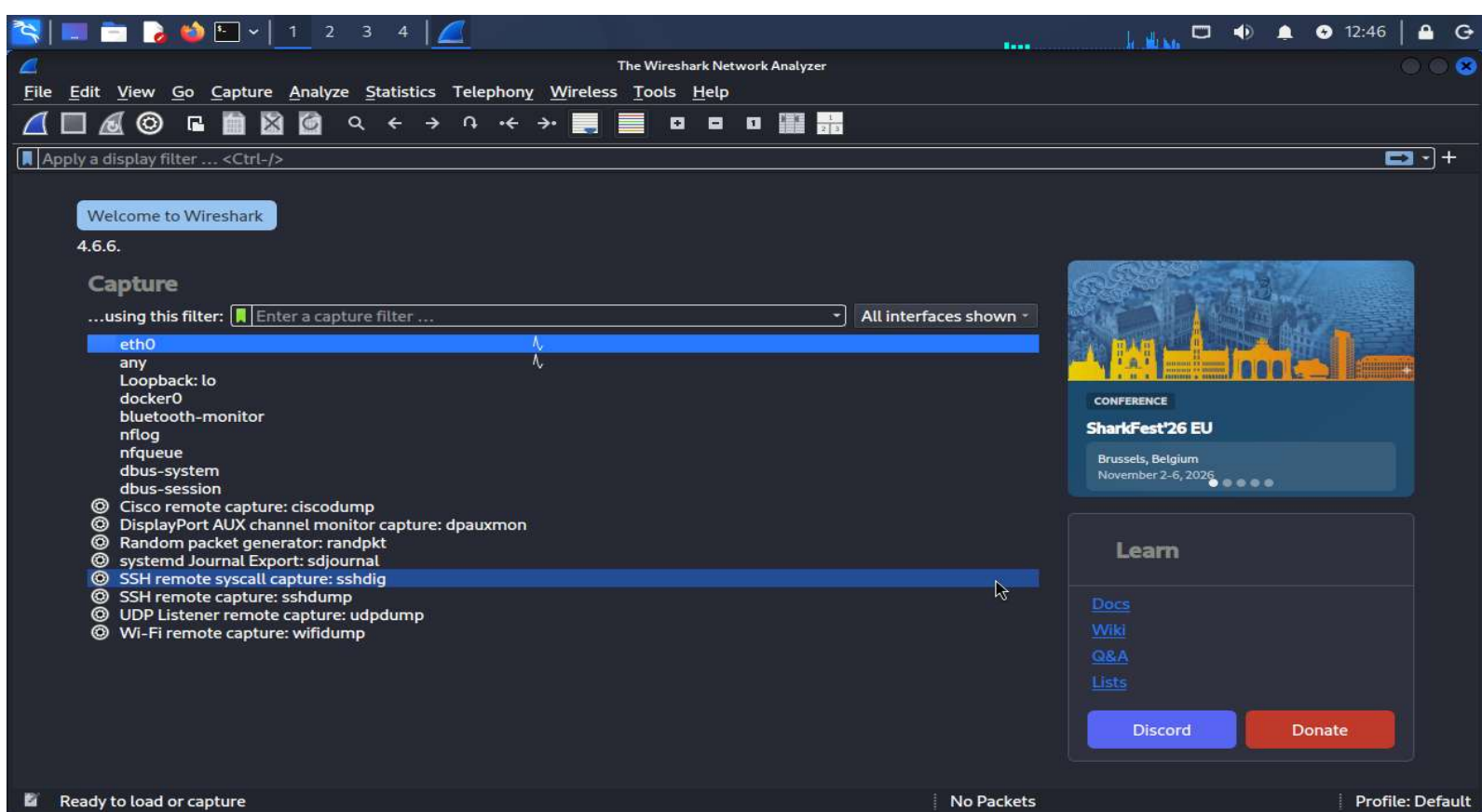


Application Protocol

This helps users understand how data moves through different layers of network communication.

8. Advantages

- Easy-to-use graphical interface.
- Supports many network protocols.
- Useful for troubleshooting.
- Helps understand network communication.
- Useful for cybersecurity education and investigations.
- Captures can be saved as **PCAP/PCAPNG** files for later analysis.

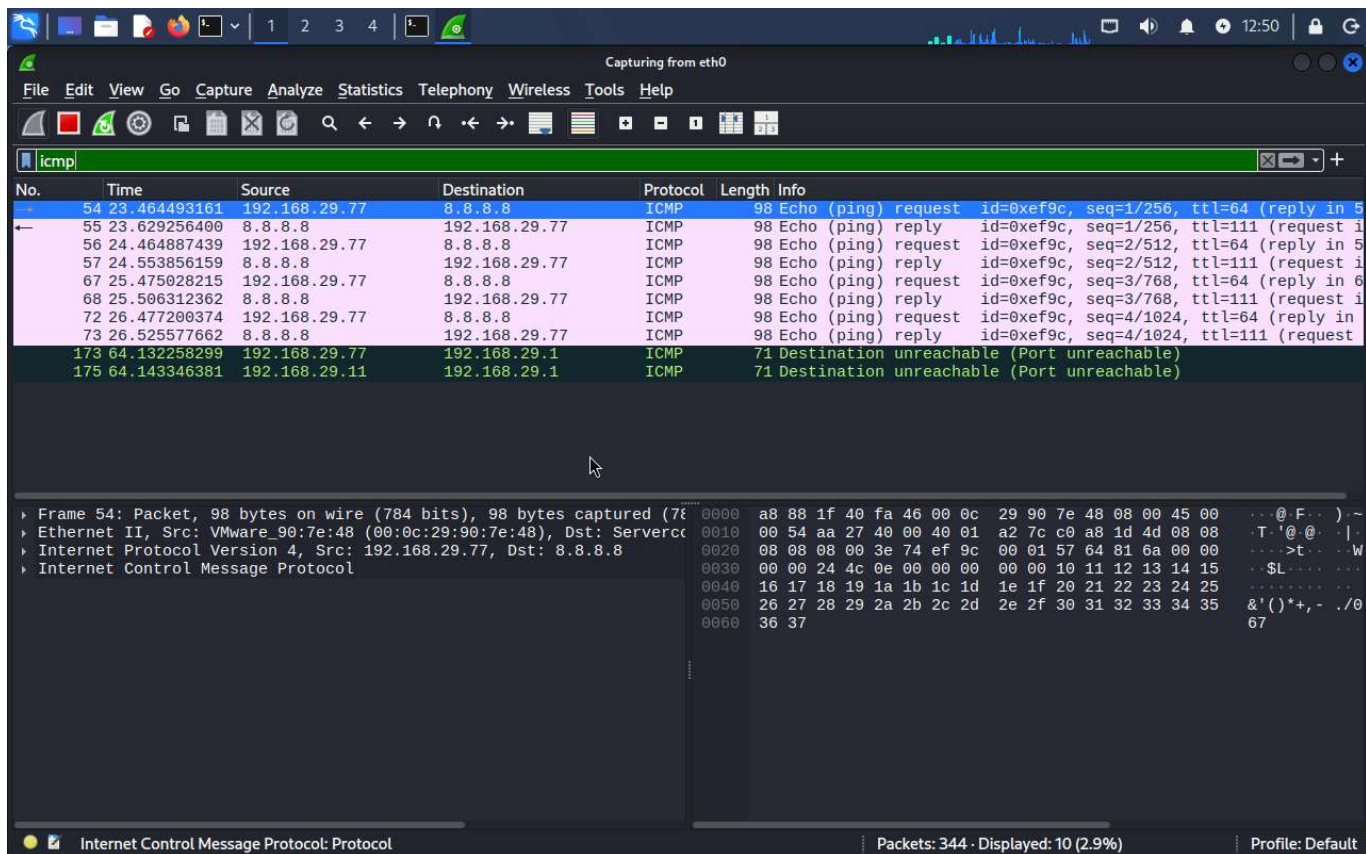


2 Nmap (network scanning).

1. Introduction

Nmap (Network Mapper) is a free and open-source tool used for network discovery and security auditing. It can identify active hosts, open ports, available services, and sometimes operating-system information.

It is commonly used by network administrators, cybersecurity students, and security professionals.



2. What is Network Scanning?

Network scanning is the process of examining a network to discover devices and services.

Scanner



Network



Devices / Hosts



Open Ports



Services



Security Analysis

3. Basic Nmap Commands

Check Nmap Installation

```
nmap --version
```

Scan a Host

Only scan a system you own or have permission to test:

```
nmap 192.168.1.10
```

This can show reachable ports and services on the target.

Scan a Local Network

For an authorized lab network:

```
nmap 192.168.1.0/24
```

This can help identify hosts on that subnet.

Service Detection

```
nmap -sV 192.168.1.10
```

The -sV option attempts to determine the **service and version** associated with discovered ports.

Operating System Detection

`nmap -O 192.168.1.10`

This attempts to identify the target's operating system. Results may require appropriate privileges and are not always accurate.

Common Port Examples

Port	Common Service
21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
443	HTTPS
3306	MySQL

5. Example Lab

Suppose your authorized lab machine has IP address:

192.168.56.101

Run:

`nmap 192.168.56.101`

You might receive output similar to:

PORT	STATE	SERVICE
------	-------	---------

22/tcp	open	ssh
--------	------	-----

80/tcp	open	http
--------	------	------

443/tcp	open	https
---------	------	-------

This tells you that ports **22**, **80**, and **443** were detected as open and associated with SSH, HTTP, and HTTPS.

6. Nmap in Cybersecurity

Nmap can be used for:

- **Network discovery**
- **Port scanning**
- **Service identification**
- **Security auditing**
- **Asset inventory**
- **Lab and penetration-testing exercises**

7. Advantages

- Free and open source.
- Works on Windows, Linux, and macOS.
- Supports many scanning techniques.
- Provides detailed network information.
- Useful for both learning and professional security audit


```
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libidn2-2.3.1 ip6v6
Compiled without:
Available nsock engines: epoll poll select

(kali@kali)-[~]
└─$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:90:7e:48 brd ff:ff:ff:ff:ff:ff
    inet 192.168.29.77/24 brd 192.168.29.255 scope global dynamic noprefixroute eth0
        valid_lft 83886sec preferred_lft 83886sec
    inet6 2405:201:2030:801f:74f2:507f:d455:6293/64 scope global temporary dynamic
        valid_lft 3585sec preferred_lft 3585sec
    inet6 2405:201:2030:801f:20c:29ff:fe90:7e48/64 scope global dynamic mngtppaddr noprefixroute
        valid_lft 3585sec preferred_lft 3585sec
    inet6 fe80::20c:29ff:fe90:7e48/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 66:2e:cb:03:db:7d brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
└─$ nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 12:55 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000014s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

(kali@kali)-[~]
└─$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 12:55 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000090s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds
```

3 Burp Suite (web proxy)

1. Introduction

Burp Suite is a web application security testing platform developed by PortSwigger. It acts as an intercepting web proxy, allowing security testers to inspect and modify HTTP/HTTPS requests and responses between a browser and a web application.

It is commonly used for authorized web application security testing.

2. What is a Web Proxy?

A web proxy acts as an intermediary between the browser and web server.

Browser



Burp Suite Proxy



Web Server



Burp Suite Proxy



Browser

This allows the tester to examine how a web application communicates with its server.

3. Main Features of Burp Suite

1. Proxy

The **Proxy** module intercepts HTTP/HTTPS traffic.

It allows you to:

- View requests and responses.
- Inspect HTTP headers.
- Examine parameters and cookies.
- Forward or drop requests.
- Modify requests in an authorized test environment.

2. Repeater

Repeater allows a captured request to be sent repeatedly while changing parameters and examining the server's responses.

It is useful for understanding how an application handles different inputs.

3. Intruder

Intruder automates customized requests for security testing, such as testing input handling and authentication controls in an authorized lab.

4. Decoder

Decoder helps encode and decode data such as:

- URL encoding
- Base64
- Hexadecimal

5. Scanner

The available **Scanner** functionality can help identify potential web application security issues, depending on the Burp Suite edition.

4. Basic Burp Suite Lab Setup

For a safe practice environment:

Step 1: Open Burp Suite.

Step 2: Open Burp's built-in browser or configure a test browser to use Burp as its proxy.

Step 3: Visit your own local test application.

For example: `http://localhost`

Step 4: In Burp, open **Proxy** → **HTTP history**.

Step 5: Select a request to inspect its:

- Method
- URL
- Headers
- Parameters
- Cookies
- Response

5. Example HTTP Request

A captured request may look like:

`GET /login HTTP/1.1`

`Host: localhost`

`User-Agent: Mozilla/5.0`

`Accept: text/html`

The server may respond:

`HTTP/1.1 200 OK`

`Content-Type: text/html`

Burp allows you to study this communication.

6. Burp Suite Workflow

Browser



HTTP/HTTPS Request



Burp Proxy



Inspect / Test



Web Server



HTTP/HTTPS Response



Burp Proxy



Browser

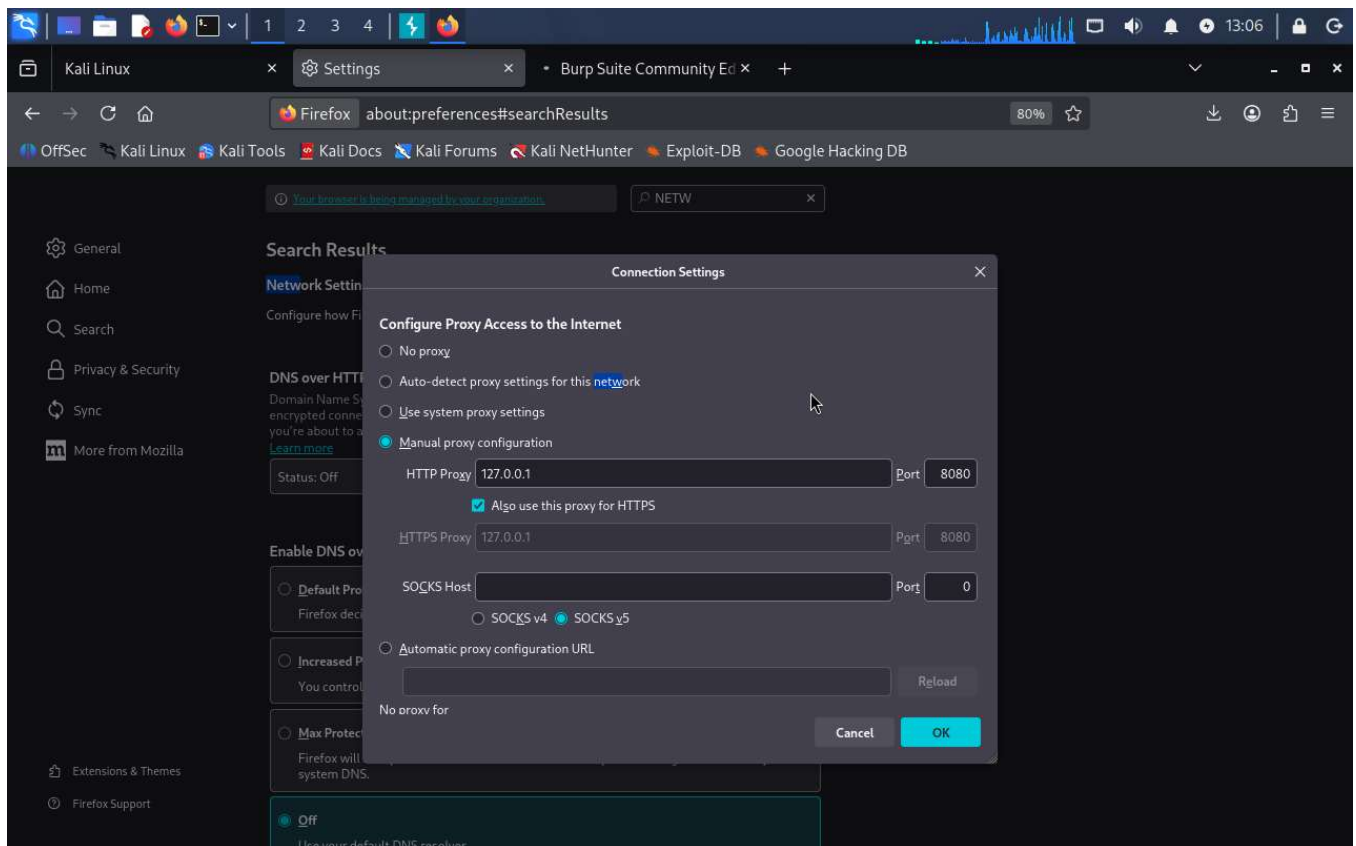
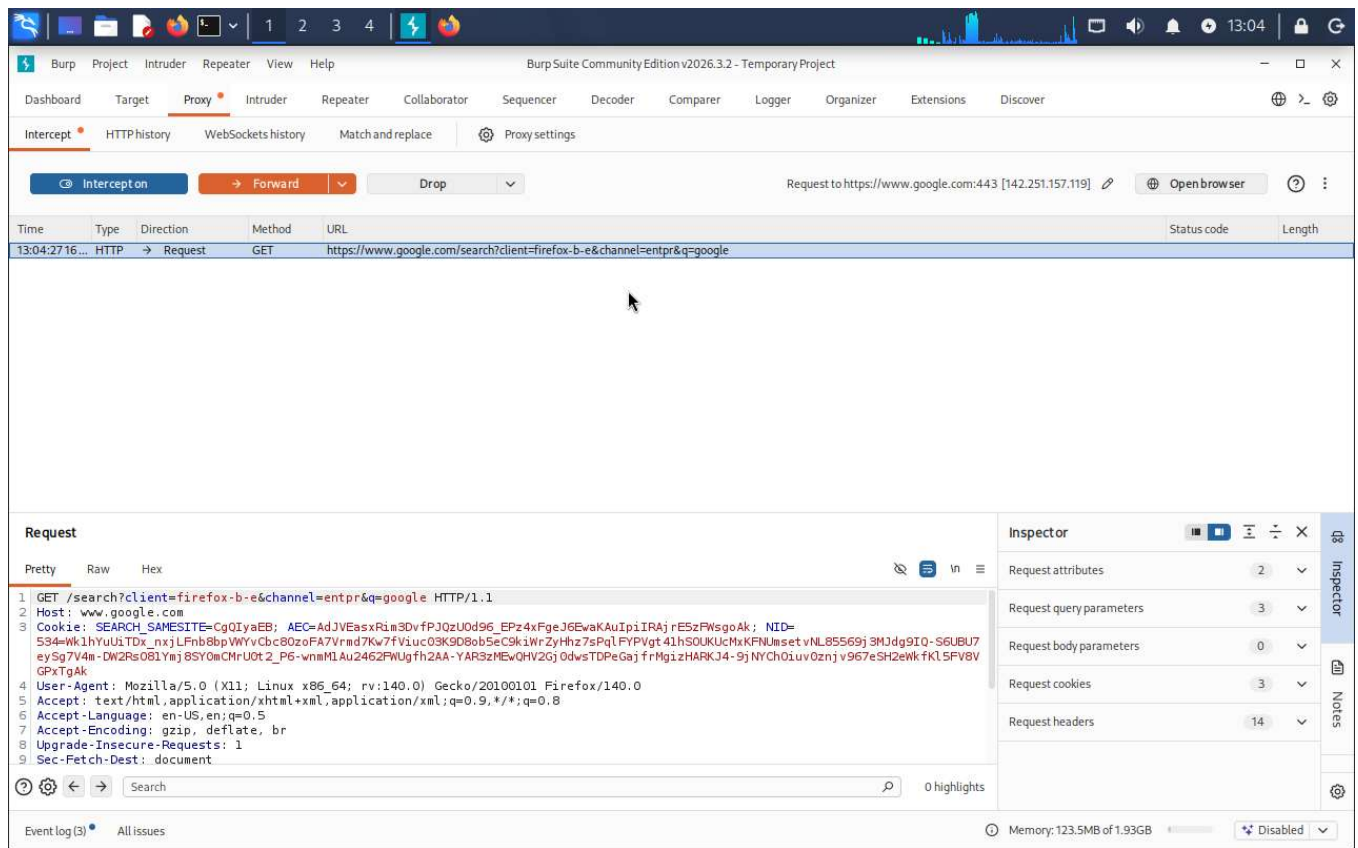
7. Uses in Cybersecurity

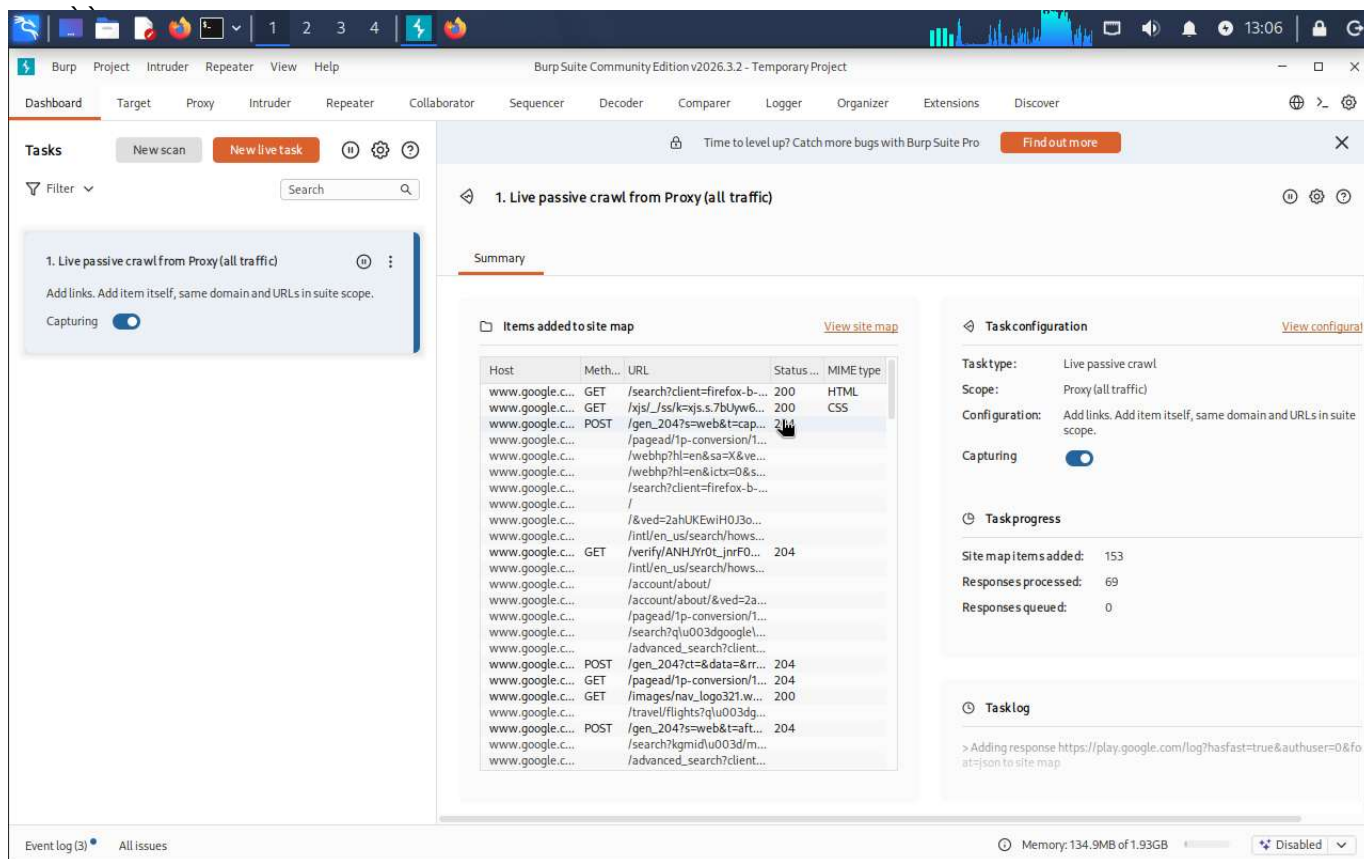
Burp Suite can help security professionals test for issues such as:

- Authentication weaknesses
- Authorization problems
- Input-validation issues
- Session-management weaknesses
- Security misconfigurations
- Common web application vulnerabilities

8. Advantages

- Powerful HTTP/HTTPS traffic inspection.
- Useful for learning web protocols.
- Provides multiple web security testing tools.
- Supports manual and automated testing.
- Widely used in professional web application security testing.





4 Netcat (network debugging)

Netcat (nc) is a command-line networking utility used to read and write data over TCP or UDP connections. It is commonly used for network debugging, testing ports, checking connectivity, and creating simple client-server connections.

1. Check if a Port Is Open

```
nc -zv 192.168.1.10 80
```

- -z → Scan without sending data
- -v → Verbose output
- 192.168.1.10 → Target IP
- 80 → Target port

Example output:

```
Connection to 192.168.1.10 80 port [tcp/http]
succeeded!
```

2. Create a Netcat Server

On **Machine 1**:

```
nc -lvnp 4444
```

- -l → Listen mode
- -v → Verbose
- -n → Don't resolve DNS
- -p 4444 → Listen on port 4444

3. Connect as a Client

On **Machine 2**:

```
nc 192.168.1.10 4444
```

Now type:

Hello from Client

The message will appear on Machine 1.

4. Simple File Transfer

Receiver: nc -lvnp 4444 > received.txt

Sender: nc 192.168.1.20 4444 < test.txt

This transfers test.txt to received.txt.

5. UDP Testing

Start a UDP listener:

```
nc -lvnup 5555
```

Connect/send UDP data:

```
nc -uv 192.168.1.10 5555
```

Practical Lab

Use **two Kali Linux VMs** on the same isolated lab network:

Kali 1 (Server)

IP: 192.168.56.10

Port: 4444



| TCP



Kali 2 (Client)

IP: 192.168.56.11

Server:

nc -lvp 4444

Client:

nc 192.168.56.10 4444

Type a message on either side and observe that it is transmitted through the TCP connection.

Important Uses

- 1.Port connectivity testing
- 2.TCP/UDP debugging
- 3.Client-server communication testing
- 4.Simple file transfer in a controlled lab
- 5.Network troubleshooting
- 6.Testing firewall rules

